

STEM Internship Learning Journal

Siu Yau Shing

Student ID: 23115372d

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Introduction

As an AI Research Engineer Intern at GBA AI Academy, my primary goal was to contribute meaningfully to the development of an AI-driven education system that transforms traditional learning and provide my programming skills to assist in their other projects. My aim was to deepen my technical expertise in AI and its implementation in systems, while gaining real-world experience in programming, system design and workplace skills.

I worked at GBA AI Academy, based in Cyberport and part of NVIDIA's Inception Program, is a startup founded by entrepreneurs. The company's core business revolves around building proprietary AI systems that analyse student engagement and emotional states to optimize learning outcomes. GBA AI Academy integrates real-time facial expression recognition (FER) to detect student focus, confusion, or frustration, enabling dynamic curriculum adjustments. Its customers include K–12 schools and online learning platforms, differentiating through localized, culturally-aware AI models and hardware-software co-design.

My internship focused on the implementation of ASR and TTS technologies into the learning platform. I learned how to implement AI models and how to use them locally. I also designed small-scale AI learning curriculums for their use in the platform. Through this work, I've honed my skills in Python, PyTorch, and AI pipelines, while contributing directly to GBA AI Academy's innovation engine: merging AI with scalable education tech.

Reflective Integrative Journal

Personal Growth

I learned a lot about real world implementation of Generative AI models and how to use them. Before the internship, I only used GenAI models using the school platform and other websites, but now I learned how it is actually deployed and implemented and how it interacts with code

such as Python, and different ways to improve its accuracy such as Finetuning and RAG, and I also had hands on experience on using them and working with them. Having practical experience building projects using programming and also doing software development and engineering have also provided me valuable experience in using these tools. As such, I have more confidence in programming in school and in my future work as well.

Impact on Future Career / Academic Planning

This internship had a deep impact on me as it made me understand the deeper workings of how GenAI models such as LLMs and other kinds of AI work and how institutions and companies use them to provide value to customers. As such, I aim to improve my understanding of how these technologies can be better integrated with current technologies and to improve my skills in programming so that I can make my own systems that integrate with AI as well. I would also aim to have a career that is related to AI, so I will focus on taking more courses related to AI as well and also develop more skills independently so I can become more competitive.

Supplementary Documentation

[GBAIA Internship Photos](#)